Past features and past labels on top of the 48-feature baseline: seven test windows

2026-11-05

This report compares the original baseline (**Run-2**: the 48 raw features of the prediction date) with models that also see each stock's recent past: the 48 features one and five trading dates earlier (T–1, T–5) and the stock's own realized labels of T–6 to T–10. All models are LightGBM models, trained anew for each rolling window on its 230 history dates; a 10-date gap separates the history from the test dates.

The models are scored on **seven consecutive test windows**, 804 test dates from 2022-12-29 to 2026-04-28, on the same dates and stocks for every column. The first six windows (W1–W6) were used while developing and choosing the setups, so their numbers flatter the chosen ones. The last window (W7, 2025-12-19 → 2026-04-28) was held back until the setups were fixed and evaluated once; Run-22 had been named the main candidate beforehand.

Table 1 describes each column (setup rows) and gives its metrics; every term is defined in the footnotes below it. Figure 1 shows the main metrics window by window. Table 2 lists the weights of the two weighted mixes.

Table 1. Setups and metrics, all seven windows

All seven windows, common rows. Below the setup rows, every metric row is computed on the same rows for all columns. Dark green (bold): best in the row at the shown precision. Light green: within the seed noise of the best. Small grey line: change vs Run-2 in % of Run-2's value. Footnotes below the table.

	Run-2 (baseline)	Run-5*	Run-11*	Run-17*	Run-22*	Run-23*	Run-11 & 17 (equal mix)*	Run-11 & 17 (weighted mix)*, **	Run-11 & 13-16 (equal mix)*	Run-11 & 13-16 (weighted mix)*, **	Seed noise [20]
Setup											
CV method [1]	Every third date	Purged 5-fold	Purged 5-fold	Purged 5-fold	Purged 5-fold	Purged 5-fold	Purged 5-fold	Purged 5-fold	Purged 5-fold	Purged 5-fold	—
Today's features used [2]	33–40	48	48	48	48	48	48	48	48	48	—
Past features used [3]	—	—	T–1: 48 T–5: 48	—	T–1: 48 T–5: 48	T–1: 48 T–5: 48	T–1: 48 T–5: 48	T–1: 48 T–5: 48	T–1: 48 T–2: 48 T–3: 48 T–4: 48 T–5: 48 T–6: 48	T–1: 48 T–2: 48 T–3: 48 T–4: 48 T–5: 48 T–6: 48	—
Past labels used [4]	—	—	—	T–6..10	T–6..10 scaled†	T–6..10 ranked††	T–6..10	T–6..10	—	—	—
Total model inputs [5]	33–40	48	144	53	149	149	144 + 53	144 + 53	5 × 144	5 × 144	—
Num. seeds [6]	1	1	1	1	1	1	1	1	1	1	—
Num. models [7]	1	1	1	1	1	1	2	2	5	5	—
Ranking, paired with Run-2 [8]											
Rank IC (mean daily) [10]	0.0592 (+0.0%)	0.0635 (+7.3%)	0.0676 (+14.3%)	0.0697 (+17.8%)	**0.0730** **(+23.4%)**	0.0729 (+23.2%)	0.0715 (+20.8%)	0.0711 (+20.3%)	0.0682 (+15.3%)	0.0683 (+15.5%)	±0.0008
ΔRank IC vs Run-2 [11]	—	+0.0043	+0.0085	+0.0105	**+0.0138**	+0.0137	+0.0123	+0.0120	+0.0090	+0.0092	±0.0008
ΔRank IC vs Run-2: 95% CI (NW, day sampling) [12], [13]	—	0.0012	0.0019	0.0046	0.0042	0.0042	0.0032	0.0035	0.0020	0.0019	—
ΔRank IC vs Run-2: t (NW) [12]	—	+6.9	+8.7	+4.4	+6.4	+6.3	+7.6	+6.8	+9.0	+9.3	—
Days better than Run-2 (%) [11]	—	64	66	60	65	64	67	67	67	**68**	±2
Windows with ΔRank IC > 0 (of 7) [11]	—	6	**7**	5	6	6	**7**	**7**	**7**	**7**	≈0

	Run-2 (baseline)	Run-5*	Run-11*	Run-17*	Run-22*	Run-23*	Run-11 & 17 (equal mix)*	Run-11 & 17 (weighted mix)*, **	Run-11 & 13-16 (equal mix)*	Run-11 & 13-16 (weighted mix)*, **	Seed noise [20]
Months with ΔRank IC > 0 (of 41) [11]	—	**35**	34	30	33	**35**	34	34	33	33	±3
Pearson IC (mean daily) [10]	0.0925 (+0.0%)	0.0960 (+3.9%)	0.0973 (+5.2%)	0.0984 (+6.4%)	0.1001 (+8.2%)	0.0998 (+8.0%)	**0.1009 (+9.2%)**	0.1007 (+9.0%)	0.0993 (+7.4%)	0.0990 (+7.1%)	±0.0007
ICIR [14]	0.956 (+0.0%)	0.989 (+3.4%)	1.029 (+7.6%)	1.026 (+7.3%)	**1.051 (+9.9%)**	1.044 (+9.2%)	1.045 (+9.3%)	1.038 (+8.5%)	1.004 (+5.0%)	1.017 (+6.3%)	±0.016
Q5 − Q1 (mean daily) [15]	0.00823 (+0.0%)	0.00872 (+5.9%)	0.00917 (+11.3%)	0.00850 (+3.2%)	0.00893 (+8.4%)	0.00901 (+9.4%)	0.00907 (+10.1%)	0.00898 (+9.0%)	**0.00927 (+12.6%)**	0.00925 (+12.3%)	±0.00013
Q1 mean label (lower = better) [15]	-0.00580 (+0.0%)	-0.00611 (-5.3%)	-0.00630 (-8.5%)	-0.00596 (-2.7%)	-0.00614 (-5.7%)	-0.00619 (-6.6%)	-0.00627 (-8.1%)	-0.00625 (-7.7%)	-0.00637 (-9.8%)	**-0.00641 (-10.4%)**	±0.00004
Q5 mean label [15]	0.00243 (+0.0%)	0.00261 (+7.5%)	0.00287 (+18.1%)	0.00254 (+4.4%)	0.00279 (+14.8%)	0.00282 (+16.2%)	0.00280 (+15.1%)	0.00272 (+12.2%)	**0.00290 (+19.2%)**	0.00284 (+17.0%)	±0.00011
Prediction agreement with Run-2 [16]	—	0.883	0.746	0.728	0.664	0.667	0.770	0.770	0.771	0.765	—
Ranking, paired with Run-5 [8]											
ΔRank IC vs Run-5 [11]	-0.0043	—	+0.0042	+0.0062	**+0.0095**	+0.0094	+0.0080	+0.0077	+0.0047	+0.0049	±0.0009
ΔRank IC vs Run-5: 95% CI (NW, day sampling) [12], [13]	0.0012	—	0.0013	0.0047	0.0041	0.0042	0.0030	0.0033	0.0014	0.0013	—
ΔRank IC vs Run-5: t (NW) [12]	-6.9	—	+6.1	+2.6	+4.5	+4.4	+5.2	+4.5	+6.8	+7.2	—
Windows with ΔRank IC > 0 (of 7) [11]	1	—	**7**	6	6	6	6	6	**7**	**7**	≈0
Rank IC per window [10]											
W1 (2022-12-29 → 2023-06-30)	0.0450 (+0.0%)	0.0485 (+7.8%)	0.0514 (+14.2%)	0.0564 (+25.3%)	0.0590 (+31.1%)	**0.0597 (+32.8%)**	0.0553 (+22.9%)	0.0551 (+22.4%)	0.0514 (+14.3%)	0.0517 (+14.9%)	±0.0012
W2 (2023-07-03 → 2023-12-25)	0.0590 (+0.0%)	0.0655 (+11.0%)	0.0668 (+13.1%)	0.0802 (+35.9%)	0.0795 (+34.8%)	**0.0808 (+36.9%)**	0.0772 (+30.8%)	0.0777 (+31.6%)	0.0686 (+16.3%)	0.0682 (+15.6%)	±0.0008
W3 (2023-12-26 → 2024-06-27)	0.0900 (+0.0%)	0.0930 (+3.4%)	0.0947 (+5.3%)	0.0868 (-3.5%)	0.0890 (-1.0%)	0.0894 (-0.7%)	0.0924 (+2.7%)	0.0913 (+1.5%)	**0.0955 (+6.2%)**	0.0954 (+6.0%)	±0.0010
W4 (2024-06-28 → 2024-12-23)	0.0556 (+0.0%)	0.0571 (+2.7%)	0.0610 (+9.8%)	0.0619 (+11.4%)	**0.0668 (+20.2%)**	0.0650 (+16.9%)	0.0645 (+16.2%)	0.0644 (+16.0%)	0.0610 (+9.9%)	0.0621 (+11.9%)	±0.0013
W5 (2024-12-24 → 2025-06-25)	0.0520 (+0.0%)	0.0574 (+10.3%)	0.0673 (+29.5%)	0.0727 (+39.8%)	**0.0780 (+50.1%)**	0.0773 (+48.7%)	0.0749 (+44.1%)	0.0744 (+43.1%)	0.0682 (+31.2%)	0.0683 (+31.3%)	±0.0019
W6 (2025-06-26 → 2025-12-18)	0.0469 (+0.0%)	0.0572 (+22.1%)	0.0632 (+35.0%)	0.0621 (+32.5%)	**0.0686 (+46.4%)**	0.0677 (+44.4%)	0.0650 (+38.7%)	0.0646 (+37.9%)	0.0629 (+34.2%)	0.0627 (+33.9%)	±0.0033
W7 (2025-12-19 → 2026-04-28), final	0.0684 (+0.0%)	0.0665 (-2.9%)	0.0694 (+1.4%)	0.0669 (-2.3%)	0.0685 (+0.0%)	0.0691 (+1.0%)	**0.0707 (+3.3%)**	0.0702 (+2.6%)	0.0702 (+2.6%)	0.0705 (+3.0%)	±0.0018
Rank IC per ex-ante regime [9]											
calm (314 dates, 10 episodes)	0.0468 (+0.0%)	0.0505 (+8.0%)	0.0543 (+16.1%)	0.0595 (+27.1%)	0.0619 (+32.3%)	**0.0626 (+33.8%)**	0.0592 (+26.6%)	0.0591 (+26.4%)	0.0541 (+15.6%)	0.0540 (+15.3%)	±0.0008
normal (262 dates, 12 episodes)	0.0519 (+0.0%)	0.0579 (+11.6%)	0.0634 (+22.1%)	0.0664 (+28.0%)	**0.0696 (+34.1%)**	0.0692 (+33.4%)	0.0681 (+31.2%)	0.0679 (+30.9%)	0.0641 (+23.6%)	0.0645 (+24.3%)	±0.0011
turbulent (228 dates, 4 episodes)	0.0845 (+0.0%)	0.0877 (+3.8%)	0.0909 (+7.5%)	0.0876 (+3.6%)	0.0922 (+9.0%)	0.0912 (+7.9%)	0.0922 (+9.1%)	0.0914 (+8.1%)	0.0923 (+9.1%)	**0.0926 (+9.5%)**	±0.0007
Magnitudes, all windows pooled [17]											
Calibration slope (best = closest to 1) [18]	0.821	0.967	**0.999**	0.916	0.987	0.992	1.025	1.014	1.022	1.010	±0.021

	Run-2 (baseline)	Run-5*	Run-11*	Run-17*	Run-22*	Run-23*	Run-11 & 17 (equal mix)*	Run-11 & 17 (weighted mix)*, **	Run-11 & 13-16 (equal mix)*	Run-11 & 13-16 (weighted mix)*, **	Seed noise [20]
Mean |slope − 1| over the 7 windows [18]	0.155	0.180	0.185	0.152	0.154	**0.146**	0.195	0.190	0.214	0.213	±0.010
R^2 [19]	0.00740 (+0.0%)	0.00896 (+21.1%)	0.00947 (+28.0%)	0.00934 (+26.3%)	0.01009 (+36.5%)	0.01008 (+36.3%)	**0.01013 (+36.9%)**	0.01008 (+36.3%)	0.00974 (+31.7%)	0.00966 (+30.6%)	±0.00011
RMSE [19]	0.05640 (+0.0%)	0.05636 (-0.1%)	0.05634 (-0.1%)	0.05634 (-0.1%)	**0.05632 (-0.1%)**	**0.05632 (-0.1%)**	**0.05632 (-0.1%)**	**0.05632 (-0.1%)**	0.05633 (-0.1%)	0.05634 (-0.1%)	≈0
R^2 per window [19]											
W1 (2022-12-29 → 2023-06-30)	0.00735 (+0.0%)	0.00904 (+22.9%)	0.00973 (+32.4%)	0.00894 (+21.6%)	**0.00977 (+32.8%)**	0.00956 (+30.1%)	0.00969 (+31.8%)	0.00972 (+32.2%)	0.00968 (+31.6%)	0.00975 (+32.6%)	±0.00076
W2 (2023-07-03 → 2023-12-25)	0.01080 (+0.0%)	0.01102 (+2.0%)	0.01183 (+9.5%)	**0.01365 (+26.4%)**	0.01340 (+24.0%)	0.01358 (+25.7%)	0.01338 (+23.8%)	0.01343 (+24.3%)	0.01240 (+14.7%)	0.01227 (+13.6%)	±0.00034
W3 (2023-12-26 → 2024-06-27)	0.01694 (+0.0%)	0.01892 (+11.7%)	0.01987 (+17.2%)	0.02009 (+18.6%)	**0.02081 (+22.8%)**	**0.02081 (+22.8%)**	0.02074 (+22.4%)	0.02071 (+22.3%)	0.02023 (+19.4%)	0.02010 (+18.6%)	±0.00033
W4 (2024-06-28 → 2024-12-23)	0.00363 (+0.0%)	0.00655 (+80.1%)	0.00693 (+90.8%)	0.00493 (+35.8%)	0.00720 (+98.0%)	0.00724 (+99.2%)	**0.00738 (+102.9%)**	0.00720 (+98.1%)	0.00719 (+97.8%)	0.00692 (+90.4%)	±0.00021
W5 (2024-12-24 → 2025-06-25)	0.00227 (+0.0%)	0.00443 (+95.1%)	0.00470 (+107.1%)	0.00520 (+128.9%)	**0.00601 (+164.8%)**	0.00597 (+163.0%)	0.00563 (+147.9%)	0.00553 (+143.4%)	0.00497 (+118.7%)	0.00499 (+119.6%)	±0.00016
W6 (2025-06-26 → 2025-12-18)	0.00640 (+0.0%)	0.00731 (+14.3%)	0.00768 (+20.1%)	0.00784 (+22.5%)	0.00784 (+22.5%)	0.00779 (+21.7%)	**0.00822 (+28.5%)**	0.00821 (+28.3%)	0.00819 (+28.0%)	0.00813 (+27.1%)	±0.00034
W7 (2025-12-19 → 2026-04-28), final	0.00568 (+0.0%)	0.00558 (-1.7%)	0.00580 (+2.2%)	0.00613 (+8.0%)	0.00610 (+7.4%)	0.00609 (+7.3%)	0.00629 (+10.9%)	**0.00631 (+11.2%)**	0.00580 (+2.1%)	0.00579 (+2.1%)	±0.00041
Calibration slope per window (best = closest to 1) [18]											
W1 (2022-12-29 → 2023-06-30)	**1.049**	1.230	1.378	1.221	1.193	1.230	1.379	1.384	1.432	1.430	±0.086
W2 (2023-07-03 → 2023-12-25)	1.127	1.121	**0.988**	1.102	1.018	1.038	1.101	1.105	1.085	1.055	±0.030
W3 (2023-12-26 → 2024-06-27)	0.945	1.060	1.033	**1.010**	1.101	1.041	1.063	1.056	1.076	1.084	±0.057
W4 (2024-06-28 → 2024-12-23)	0.633	0.748	0.801	0.648	0.772	**0.802**	0.789	0.773	0.789	0.762	±0.036
W5 (2024-12-24 → 2025-06-25)	0.596	0.755	0.817	0.820	0.825	0.846	**0.898**	0.893	0.832	0.839	±0.043
W6 (2025-06-26 → 2025-12-18)	**0.946**	1.163	1.299	1.071	1.162	1.145	1.262	1.222	1.335	1.328	±0.048
W7 (2025-12-19 → 2026-04-28), final	**1.027**	1.191	1.195	1.127	1.198	1.215	1.247	1.227	1.192	1.196	±0.053
Cost [21]											
Training time, all windows (minutes)	**3.2**	12.6	14.4	12.3	15.0	14.4	26.7	26.7	72.9	72.9	—
Mean trees per model and window	803	402	381	388	381	373	384	384	388	388	—

Footnotes

Column marks

- * **Demeaned predictions.** On every test date, the average of the model's predictions over all stocks that day is subtracted from each stock's prediction: demeaned prediction = prediction − (mean prediction of all stocks on that date). This uses only that date's predictions, no labels, so it can be done when the predictions are made. It doesn't change the order of stocks within a date, so Rank IC, Pearson IC, Q1/Q5 and agreement are unchanged. It does change the size of the predictions, and with it R^2, RMSE and the calibration slope: the label averages about zero on every date, and the models' daily average prediction doesn't follow the label's daily average. In a mix, every model's predictions are demeaned before mixing (so the mix is demeaned too). Run-2 is scored on its raw predictions, as the author trained it.
- ** **Weighted mix: how the weights are obtained.** Separately for each window, from that window's 230 history dates only (never its test dates):

  1. Each model of the mix was trained with purged 5-fold cross-validation [1]. Every history date falls in exactly one of the five validation blocks and is predicted by the fold model that did not train on that block, at the tree count the final model uses. So

every history row has one prediction from a model that never saw it (an "out-of-fold" prediction).
2. These out-of-fold predictions are demeaned per date, as in *.
3. On the history rows that every model of the mix predicts, the weights w_1 … w_k (k = 2 or 5) minimize Σ over rows of (label − $w_1 \cdot p_1$ − … − w_k·p_k)², under w_j ≥ 0 and w_1 + … + w_k = 1. This is a least-squares fit with constraints, solved numerically (SciPy's SLSQP method). A weight can come out exactly 0, which drops that model.
4. The window's test predictions are mixed with those weights: mix = $w_1 \cdot p_1$ + … + w_k·p_k (each p_j demeaned). Table 2 lists the weights of every window.

The equal mixes use w_j = 1/k and fit nothing. W7's weights are those of the final evaluation. W1–W6's were fitted for this report, after the final evaluation.

Setup rows

- **[1] CV method:** how each model chooses its number of trees on its 230 history dates.
  - *Every third date* (Run-2, the author's method): history dates 1, 4, 7, … are set aside for validation and the model trains on the other two thirds. Trees are added until a validation score (RMSE or R^2) hasn't improved for 200 trees, and the model at the best tree count predicts the test dates directly, without retraining. A label covers about the next 5 days, so a validation date's label overlaps those of its neighbouring training dates, which flatters validation.
  - *Purged 5-fold*: the 230 history dates are cut into 5 consecutive blocks of 46 dates. Five fold models each validate on one block and train on the remaining dates, minus the 10 dates on each side of the block (the "purge", so no training label overlaps a validation label). The five are grown tree by tree together; the tree count is the one with the highest average of the five validation R^2 values (stopping once it hasn't improved for 200 trees). A final model is then trained on all 230 history dates with that tree count, and it predicts the test dates.
  - Every window then leaves a 10-date gap between its last history date and its first test date, so no training label overlaps a test label.
- **[2] Today's features used:** how many of the 48 raw features of the prediction date T the model takes. Run-2 keeps only the features whose correlation with the label over the history dates exceeds 0.02 in absolute value, so the count changes per window (the range over the seven windows is shown). All other setups use all 48.
- **[3] Past features used:** "T–n: X" means the model also takes the values of X raw features from the stock's n-th previous trading date (missing if the stock has no row that date; the model handles missing values on its own). For the mixes, the rows list every lag that at least one of its models uses. In the 11 & 13-16 mixes, each model takes T–1 and exactly one of T–2, T–3, T–4, T–5, T–6 (runs 13, 14, 15, 11 and 16), so each model has 144 inputs.
- **[4] Past labels used:** "T–6..10" means five inputs: the stock's own labels from the 6th, 7th, 8th, 9th and 10th previous trading dates. A label covers about the next 5 days, so on date T only labels of dates T–6 and earlier are known. These inputs are missing when the stock has no label on that date. In the 11 & 17 mixes only run 17's model takes them.
  - **† scaled:** each past label is divided by the standard deviation of all stocks' labels on that label's own date (date T–n, not T). This removes market-wide changes in the size of labels (e.g. after a shock) and keeps the stock's sign and its position relative to the other stocks.
  - **†† ranked:** each past-label input is replaced by its percentile among all stocks on the prediction date T: its rank (1 = smallest; tied values share their average rank) divided by the number of stocks with a value that date, which gives a number between 0 and 1. Missing values stay missing.
- **[5] Total model inputs:** today's features + past features + past labels, per model. For mixes, per model: "144 + 53" = run 11's model and run 17's model; "5 × 144" = five models of 144 inputs each.
- **[6] Num. seeds:** every model is trained once per window, with one random seed (the seed of LightGBM's row sampling). Other seeds of runs 5, 11 and 22 exist for W1–W6; they are used only for the Seed noise column [20].
- **[7] Num. models:** how many separately trained models are averaged into one prediction per window.

Metric rows

- **[8] Common rows, paired comparison:** in every window, only the (date, stock) rows on which every column has a prediction are used, so all columns are scored on exactly the same rows (Run-2's own row filter keeps slightly different rows from the other setups). "Paired" means a column and its reference are compared date by date on those same rows.
- **[9] Ex-ante regime:** a label for each test date, computed only from information known on that date. For date T: the standard deviation across stocks of the labels of each of the 20 dates T–26 … T–7 (all known by T), averaged ("trailing dispersion"). It is then compared with the same quantity on the 250 dates before T: the share of those past values that are lower gives a percentile. Calm: percentile below 1/3; normal: 1/3 to 2/3; turbulent: 2/3 or above. "Episodes" counts stretches of at least 5 consecutive test dates in the same regime: neighbouring dates are alike, so the number of episodes, not of dates, tells how much independent evidence a regime holds. Labels are already adjusted for common market factors, so this measures how widely individual stocks' labels spread, not market volatility.
- **[10] Rank IC, Pearson IC:** on each test date, the correlation across stocks between the prediction and the label: Spearman (correlation of the ranks) for Rank IC, ordinary (Pearson) correlation for Pearson IC. The table shows the average over all test dates (804 for all windows). Pearson IC gives more weight to stocks with extreme labels; Rank IC only uses the order.
- **[11] ΔRank IC, days, windows and months better:** for each test date, d = (the column's Rank IC) − (the reference's Rank IC) on the same rows. "ΔRank IC" is the average of d over all dates; "days better" the share of dates with d > 0; "windows" and "months" count the test windows and the calendar months in which the average of d is above 0 (months with only a few test dates, e.g. the first and last of a window, count like full ones).
- **[12] t-tests (NW = Newey–West):** each test is a test of whether the average daily difference d [11] is zero. Step by step:
  1. Compute d for each of the T = 804 test dates, in date order (all windows; the windows follow each other without a break). Pairing removes what moves both models' IC on the same date (e.g. a day on which every model does well).
  2. The average $\bar{d}$ = (d_1 + … + d_T) / T.

3. Its standard error, allowing for correlation between nearby dates. A label covers about 5 days, so labels of neighbouring dates overlap and d values of nearby dates are correlated (daily ICs: about 0.5–0.7 from one date to the next). A plain t-test assumes independent dates and would overstate t by about 1.7×. The Newey–West estimate: with $e_t = d_t - \bar{d}$ and $\gamma_j = (1/T)\cdot\Sigma_{t>j}\ e_t\cdot e_{t-j}$ (the covariance of e with itself j dates earlier), $\mathrm{Var}(\bar{d}) = [\gamma_0 + 2\cdot\Sigma_{j=1..4}\ (1 - j/5)\cdot\gamma_j] / T$ and $SE = \sqrt{\mathrm{Var}(\bar{d})}$. It uses 4 lags because the overlap spans about 5 dates; the weights $1 - j/5$ shrink linearly with the distance, which keeps the variance positive.
4. $t = \bar{d} / SE$. The 95% confidence interval is $\bar{d} \pm 1.96\cdot SE$, and the table shows the half-width $1.96\cdot SE$. $|t| > 1.96$ is the same as an interval that excludes 0 (two-sided, 5% level, normal approximation; T is large).

The tests aren't corrected for multiple comparisons: ten columns are shown, and many more variants were tried on W1–W6.

- **[13] Day sampling:** the confidence interval covers only the randomness of which dates happen to be in the test period, with the trained models taken as fixed. It does not include the randomness of training (the seed): that is the Seed noise column [20].
- **[14] ICIR:** average daily Rank IC divided by its standard deviation across dates (how steady the ranking skill is from date to date; not scaled by the number of dates).
- **[15] Q1, Q5, Q5 − Q1:** on each test date, the stocks are sorted by the column's prediction and cut into five equal groups (quintiles); Q1 is the fifth with the lowest predictions, Q5 the fifth with the highest. The table shows each group's average label, averaged over dates, and the difference Q5 − Q1. This is a gross spread in label units, before trading costs and ignoring that labels overlap and that some stocks can't be traded or sold short; it isn't a portfolio return.
- **[16] Prediction agreement:** on each test date, the Spearman correlation between the column's predictions and Run-2's, averaged over dates (1 = the same order of stocks).
- **[17] Magnitudes:** these rows judge the size of the predictions, not only their order, pooled over all common rows of all windows. Run-2 is raw; the other columns are demeaned (*).
- **[18] Calibration slope:** the slope of a straight-line fit of the label on the prediction over the rows, cov(prediction, label) / var(prediction). 1 = predictions of the right size; below 1 = too large (overconfident); above 1 = too small. The pooled slope can look right while windows err in opposite directions, so the mean of |slope − 1| over the seven windows is shown too.
- **[19] R^2, RMSE:** $R^2 = 1 - \Sigma(\text{label} - \text{prediction})^2 / \Sigma(\text{label} - \text{average label})^2$, over the rows (the average over the same rows). 0 = no better than predicting the average label for every stock; it can be negative. RMSE = √(average of (label − prediction)²), in label units. Both use the predictions as they are, without any rescaling.
- **[20] Seed noise:** how much a metric moves when only the random seed of training changes. Runs 5, 11 and 22 were trained with three seeds each on W1–W6 (nine seed runs; W7 has one seed per setup). For each row, the metric is computed for the nine seed runs on W1–W6. σ^2 is the variance within each setup's three seeds, pooled over the three setups (sum of squared deviations from each setup's mean, divided by 9 − 3). The column shows $\pm 1.96\cdot\sqrt{2}\cdot\sigma$: the range that holds 95% of the differences between two runs that differ only in their seed. Two columns closer than this can't be told apart by these seed runs.
  - W7 rows (no seed variants) use the per-window noise of W1–W6, pooled over the six windows.
  - Rows compared with Run-5 use only runs 11 and 22 (six seed runs, σ^2 divided by 6 − 2): Run-5's other seeds would be compared with Run-5 itself.
  - Rows over all windows use the W1–W6 values (720 dates), so they slightly overstate the noise for 804 dates.
  - It assumes that every setup is about as sensitive to the seed as runs 5, 11 and 22. Mixes average several models, so they are probably less sensitive. With three seeds per setup, σ is only a rough estimate.
  - Light green in the table marks values within this band of the row's best.
- **[21] Cost:** training time is the sum over the seven windows of the recorded time to train each model (fold models and the final model), in minutes, on the machine the run used. Mixes add up all their models. Trees are the final models' tree counts, averaged over windows (and over a mix's models).
- **% line under a value:** (value − Run-2's value) / |Run-2's value|, in %.

Figure 1. Per window: ranking, magnitudes and calibration

Four panels, one line per column, the final window W7 shaded. Run-2 is black. In each panel, the **winner** of that aspect is drawn with a thicker line and named in bold in the legend (and under the panel title). The winner is judged over all seven windows together, as in the grand table:

- Rank IC, Pearson IC: the highest average over all 804 test dates;
- R^2: the highest R^2 pooled over all common rows;
- Calibration: the lowest mean |slope − 1| over the seven windows (the slope closest to 1 on average; the dotted line marks 1).

R^2 and calibration use demeaned predictions (*), except Run-2 (raw).

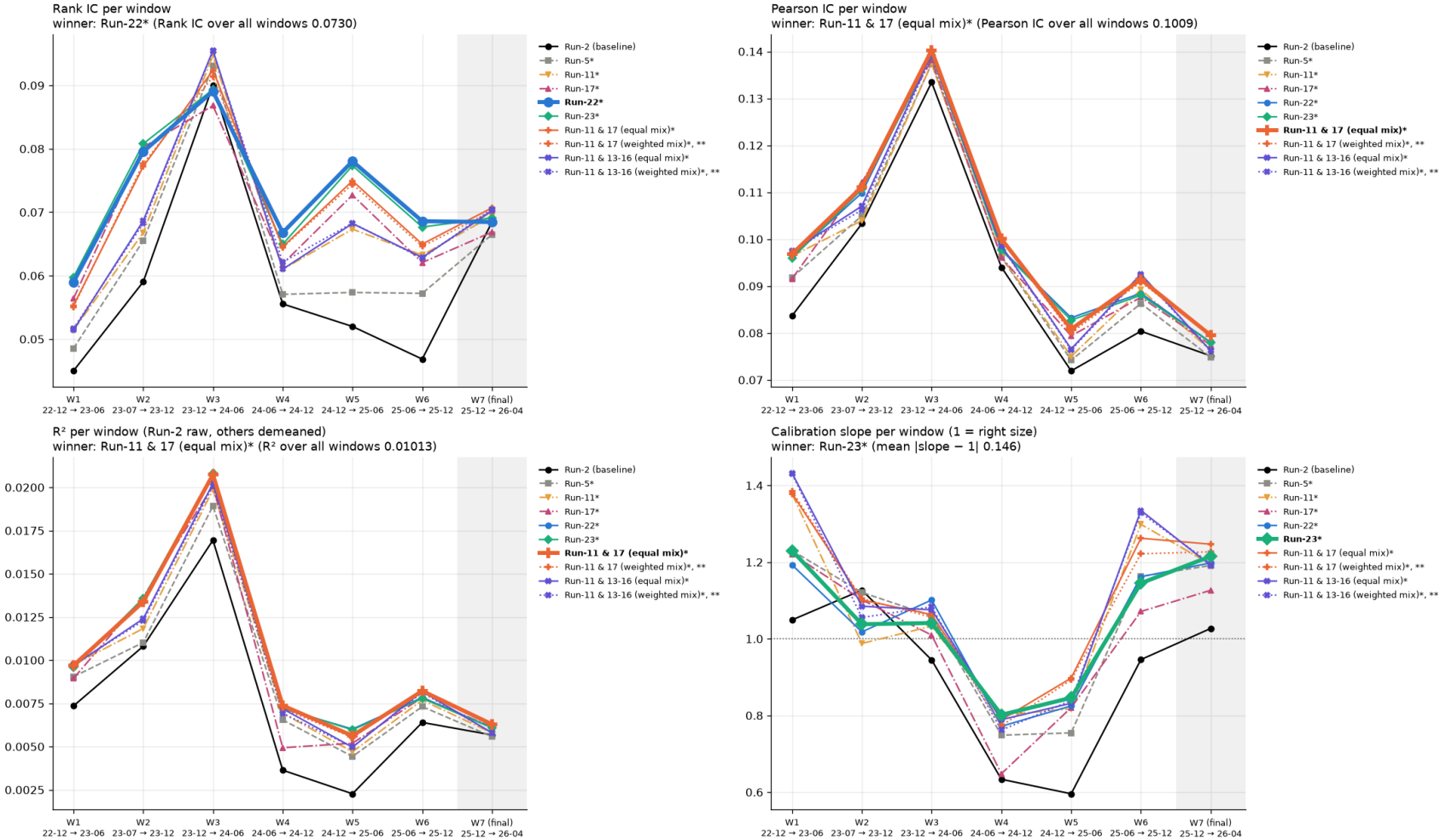


Table 2. Weights of the weighted mixes, per window

Fitted on each window's history only (footnote **); a weight of 0 drops that model in that window.

Mix	Model	W1	W2	W3	W4	W5	W6	W7 (final)
Run-11 & 17 (weighted mix)	Run-11	0.534	0.469	0.363	0.428	0.623	0.371	0.357
Run-11 & 17 (weighted mix)	Run-17	0.466	0.531	0.637	0.572	0.377	0.629	0.643
Run-11 & 13-16 (weighted mix)	Run-11	0.301	0.422	0.275	0.000	0.000	0.268	0.381
Run-11 & 13-16 (weighted mix)	Run-13	0.328	0.108	0.172	0.702	0.520	0.101	0.357
Run-11 & 13-16 (weighted mix)	Run-14	0.000	0.112	0.000	0.283	0.099	0.352	0.057
Run-11 & 13-16 (weighted mix)	Run-15	0.328	0.284	0.000	0.000	0.070	0.168	0.000
Run-11 & 13-16 (weighted mix)	Run-16	0.043	0.074	0.552	0.015	0.311	0.111	0.205

Summary

- **Ranking.** Adding the 48 features of T–1 and T–5 and the stock's labels of T–6 to T–10 (Run-22) raises the mean daily Rank IC from 0.0592 (Run-2) to 0.0730 over the 804 test dates (+0.0138, t 6.4): better in 6 of 7 windows and on 65% of the dates. Against Run-5, which is trained like Run-22 but on the 48 features of the prediction date only, the added inputs account for +0.0095 (t 4.5).
- **Magnitudes.** R^2 rises from 0.00740 to 0.01009. Part of this comes from the training procedure and the demeaning, as Run-5 shows (0.00896). Calibration varies with the period for every column (slopes from about 0.6 to 1.4); averaged over the windows, Run-22's distance from a slope of 1 is the same as Run-2's (0.154 vs 0.155).
- **The two kinds of history.** The past features (Run-11) and the past labels (Run-17) each help on their own. Combined in one model (Run-22) they give the best ranking; the average of the two separate models (Run-11 & 17 mix) gives a similar R^2 and Pearson IC.
- **Dependence on the period.** The gain over Run-2 differs a lot between windows: largest in W2, W5 and W6 (about +0.02 Rank IC), close to zero in W3 and in the final window W7, the only window that played no part in choosing the setups. More recent data are needed to see whether the improvement holds across market periods.